Determining the Efficacy of Ascorbic Acid on Reducing the Sodium Quantity in F11 Cells Modeling Peripheral Neuropathy

Introduction

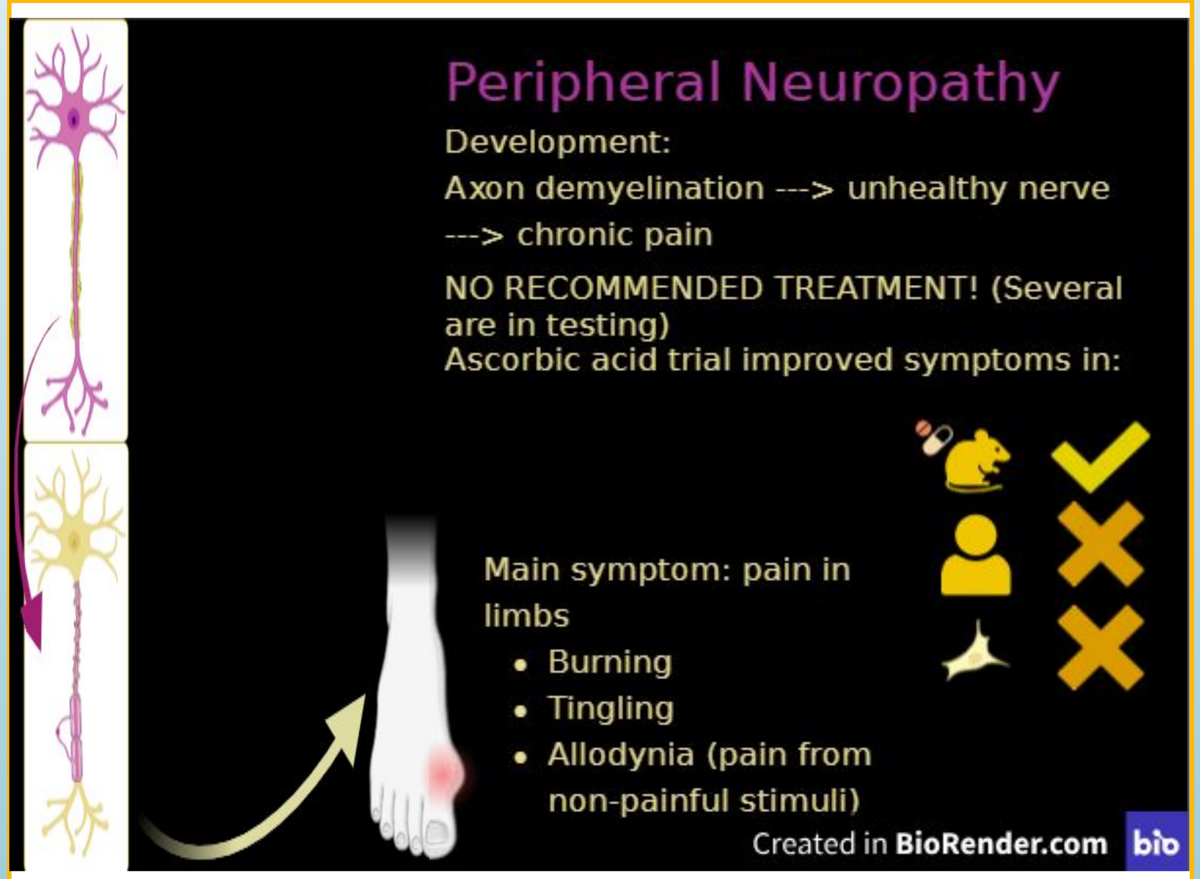


Figure 1: overview of how peripheral neuropathy develops, symptoms, and how ascorbic acid has been tested as a possible treatment in only mice.

The problem to be solved is peripheral neuropathy, an untreatable chronic condition that can cause pain. It is caused when the myelin sheath around neurons deteriorates and cannot heal, making cells sensitized to chronic pain. High sodium influx in nerve cells is associated with high levels of pain.

F11 cells are rat dorsal root ganglion, similar to human. NaVs are voltage-gated sodium channels in neurons that activate when neurons fire. Because this project is generalizable to human nerves, it will represent the probability of a common drug helping to treat pain in hundreds of thousands of people.

It is hypothesized that increasing the concentration of ascorbic acid (AA) administered to F11 cells will result in lower activity in NaVs, and therefore a lower expression of pain, because ascorbic acid has mitigated chronic pain in mice.

References

Kushnarev, M., Pirvulescu, I. P., Candido, K. D., & Knezevic, N. N. (2020). Neuropathic pain: preclinical and early clinical progress with voltage-gated sodium channel blockers. *Expert Opinion on Investigational Drugs, 29*(3), 259–271. https://doi.org/10.1080/13543784.2020.1728254

Tsubota, M., Uebo, K., Miki, K., Sekiguchi, F., Ishigami, A., & Kawabata, A. (2019). Dietary ascorbic acid restriction in GNL/SMP30-knockout mice unveils the role of ascorbic acid in regulation of somatic and visceral pain sensitivity. *Biochemical and Biophysical Research Communications 511*(3), 705-710. https://doi.org/10.1016/j.bbrc.2019.02.102

Yang, K., Wang, Y., Li, Y. W., Chen, Y. G., Xing, Na., Lin, H. B., Zhou, P., & Yu, X. P. (2022). Progress in the treatment of diabetic peripheral neuropathy. *Biomedicine and Pharmacotherapy 148*. https://doi.org/10.1016/j.biopha.2022.112717

Methodology

IV: concentration of AA.

DV: fluorescence to be measured with a BioTek plate reader.

Constants: temperature of cell culture (37°C) and 5% carbon dioxide, seeding density (20,000 cells/well), time of growth (48 hrs), black well plate used to seed all cultured F11 cells.

Well plates were either seeded to be half given TNF-α and half not given TNF-α or to be all given TNF-α and then treated with AA as shown below as shown below:

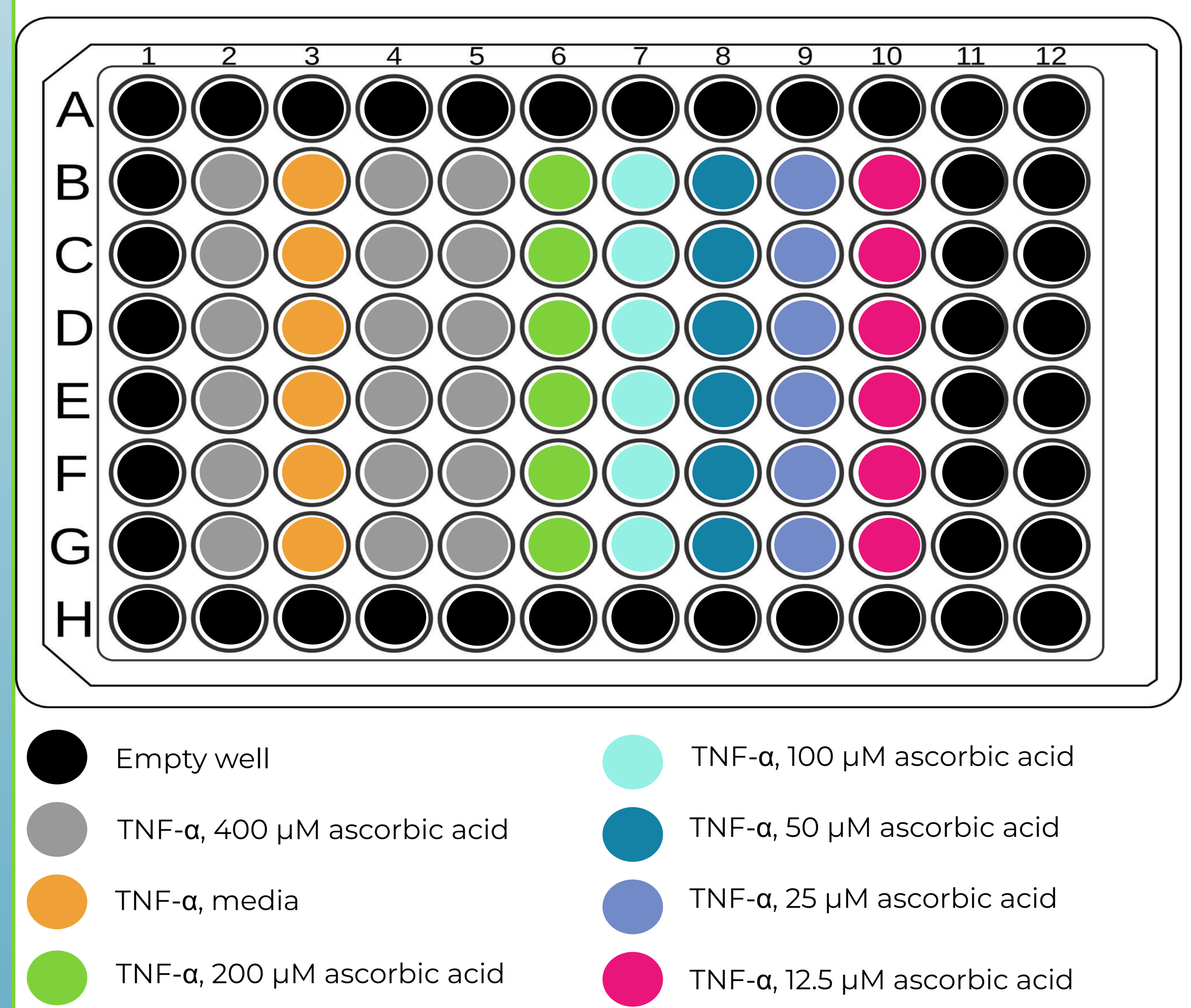
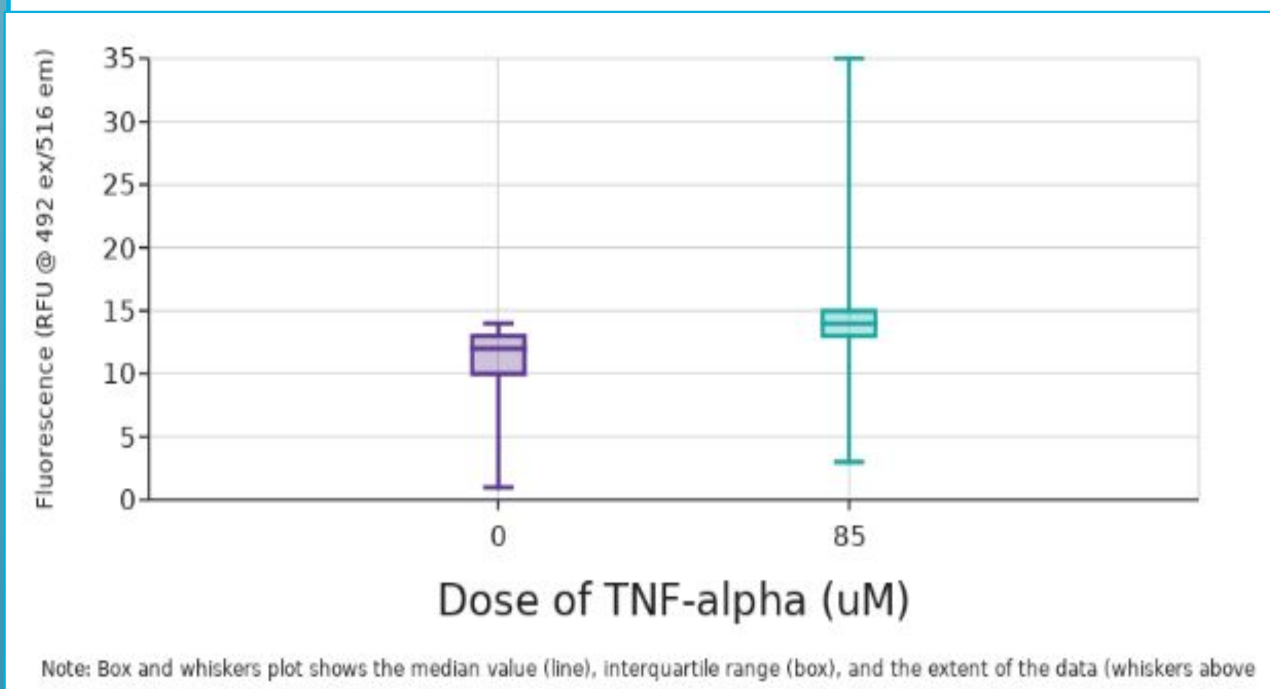


Figure 2: setup of the experimental well plates. All wells that are not empty had cells, CoroNa Green, and amaranth. Wells with no TNF-α explicitly stated did not receive it.

TNF-α, which increases neuron firing rate, is administered 48 hours after seeding and incubated with for 20 minutes. Then, AA is added to experimental cells and incubated with for another 30 mins. Finally, the fluorescence indicator CoroNa Green and excess dye quencher amaranth are added. To collect data, fluorescence is measured at 485 nm excitation with range 20 nm, 530 nm emission with range 25 nm with the BioTek plate reader. Fluorescence was measured at time = 0 minutes and at 10 minutes with incubation in between.

Data and Results



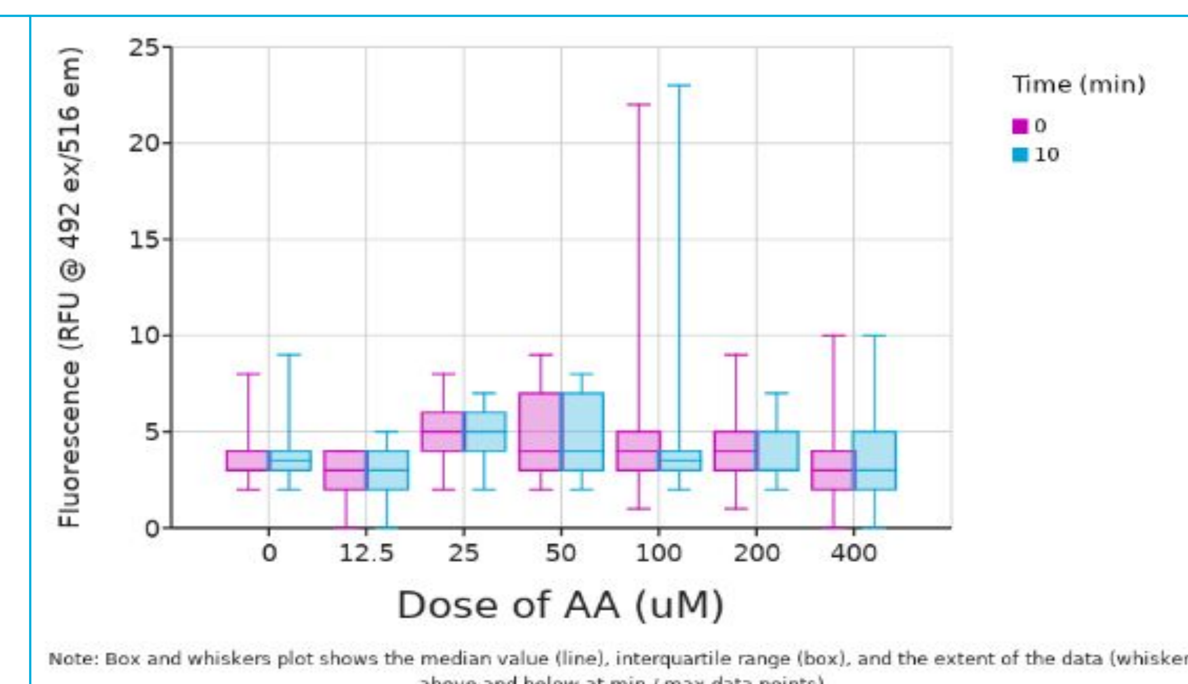


Figure 3: the fluorescence of cells given no ascorbic acid but are in the TNF-α group. Cells with TNF-α simulate inflamed cells, and cells without TNF-α mimic normal neurons.

Figure 4: fluorescence of inflamed cells treated with various doses of AA.

Conclusions

A Mann-Whitney test ($p < 0.01$), imported from Data Classroom, determined that there is a statistically significant difference between the fluorescence of cells given and not given TNF-α, with cells that do have TNF-α exhibiting higher fluorescence. A statistically significant difference is $p < 0.05$.

A Kruskal-Wallis test ($p < 0.01$), imported from Data Classroom, determined that there is a statistically significant difference between the fluorescence of cells given varying doses of AA. A Dunn's Test further revealed that the statistically significant difference ($p < 0.05$) is only between 25 μM and 50 μM, 25 μM and 100 μM, and 12.5 μM and 50 μM. Therefore, it can be concluded that there is a statistically significant difference between the sodium flux of inflamed cells treated with 25 μM and 50 μM of AA, with 50 μM being more effective at reducing sodium flux; 25 μM and 100 μM of AA, with 100 μM being more effective at reducing sodium flux; and 12.5 μM and 50 μM, with 12.5 μM being more effective at reducing sodium flux.

A Mann-Whitney test ($p = 0.97$), imported from Data Classroom, determined that there is no statistically significant difference between the fluorescence of cells at time = 0 minutes and 10 minutes.

The only supportable conclusions are: that there is no statistically significant difference in the fluorescence of cells before and after incubation with CoroNa Green. These results indicate that the no-wash CoroNa Green assay needs more development to be a reliable indicator of the change in sodium within cells over a period of time. However, TNF-α was shown to reliably induce inflammation-like activity in F11 cells. Finally, AA doses of 12.5 μM and 100 μM were more beneficial for pain derived from hyperactivity of NaVs in nerve cells than 50 μM.

The hypothesis is not supported, because a low concentration of AA was shown to have similar effects as the highest doses, while there was no statistically significant difference in fluorescence between most of the doses, and therefore no statistically significant difference in the expression of pain.

All images were created by student researcher.